Junsung Kim

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Research Interests

Topics: Vision–language models and multimodal LLMs; visual reasoning; mixture-of-experts and conditional computation; diagnostic analysis of model behavior; hyperspectral face anti-spoofing.

Current focus: How image–text interactions should condition computation in multimodal models—when learned expert weighting improves visual perception and reasoning, and when it collapses toward uniform gating.

Education

Gwangju Institute of Science and Technology (GIST) 2025.03 – 2027.02 (expected)
M.S. in Electrical Engineering and Computer Science Gwangju, South Korea
Thesis: A Study of Spatial Token–Patch Interactions and Soft Dense Expert Weighting in Vision–Language Models
Advisor: Prof. Heung-No Lee
GPA: 3.66/4.5

Ajou University 2019.03 – 2025.02
B.S. in Software and Computer Science Suwon, South Korea
B.A. in Political Science and Diplomacy (double major)
GPA: 3.7/4.5

Research Experience

INFONET Lab, Gwangju Institute of Science and Technology (GIST) 2025.03 – Present
Graduate Researcher; Advisor: Prof. Heung-No Lee Gwangju, South Korea

Research on vision–language models, mixture-of-experts architectures, hyperspectral imaging, and LLM reasoning, resulting in six papers and manuscripts (two as first author). Also co-authored work on retrieval-augmented interpretation and dataset creation for Korean folk painting (Minhwa) [3, 5] and on generative game-asset synthesis [6].

IP-MoE: Token–Patch Interactions for Expert Weighting in Vision–Language Models
First author; under review at ICLR 2027 [1]

- Proposed and implemented a vision–language connector in which bilinear token–patch interactions set the weights of a dense mixture of six spatial experts and condition the language model; integrated it with frozen CLIP encoders and Llama-3.2-3B-Instruct.
- Built data-preprocessing and two-stage distributed training pipelines on the LLaVA-NeXT dataset, using LoRA and BF16 mixed precision across six NVIDIA A100 GPUs.
- Designed controlled comparisons against additive interactions and uniform expert weighting across seven benchmarks to test whether bilinear interactions and learned weighting improve visual perception and reasoning.
- Diagnosed model behavior through routing-entropy, per-subtask, and prediction-pattern analyses, finding that expert gates become nearly uniform late in training and identifying an answer-option imbalance on CV-Bench-3D.

Evidence Minimization for Private LLM Reasoning
Co-author; under review at ICLR 2027 [2]

- Contributed to constructing the study's scientific-reasoning benchmark of 160 workflows and 800 tasks across four domains.
- Provided feedback on the scientific evidence-minimization method and algorithm, and reviewed the manuscript for the soundness of its reasoning and logical flow.

Hyperspectral Face Anti-Spoofing and Identification
First author; oral presentation at KICS Winter Conference 2026 [4]

- Extended a VGG-style CNN with four sparsely gated experts for both hyperspectral face anti-spoofing and identity recognition.
- Implemented the preprocessing, training, and evaluation pipeline on 4,374 images from 54 participants; benchmarked the model against a CNN-only baseline.

Publications

Under Review

[1] "IP-MoE: Studying spatial token–patch interactions and dense expert weighting," under review at ICLR 2027 — 2026
Junsung Kim, *Youngin Choi*, *Minjun Song*, *Sumin Kim*, *Surin Lee*, *and Heung-No Lee*

[2] "Beyond answer preservation: Scientific evidence minimization for private LLM reasoning," under review at ICLR 2027 — 2026
Surin Lee, ***Junsung Kim***, *Minjun Song*, *Sumin Kim*, *and Heung-No Lee*

Workshop Paper

[3] "Retrieval-augmented generation for symbolic folk-art interpretation: Comparing dense and graph-based retrieval using Korean Minhwa," ORACLE Workshop (Open Reasoning Across Cultures & Languages) at EMNLP 2026, Budapest, Hungary (to appear) — 2026
Minjun Song, *Sumin Kim*, ***Junsung Kim***, *Yebeen Lee*, *and Heung-No Lee*

Domestic Conference Papers (Korea)

[4] "An HSI-based CNN-MoE framework for face anti-spoofing and identification," KICS Winter Conference 2026, Pyeongchang, South Korea, pp. 204–205 (oral presentation) — 2026
Junsung Kim, *Youngin Choi*, *and Heung-No Lee*

[5] "AI-assisted dataset creation pipeline for Korean Minhwa," KICS Winter Conference 2026, Pyeongchang, South Korea, pp. 1596–1597 — 2026
Minjun Song, *Sumin Kim*, ***Junsung Kim***, *and Heung-No Lee*

[6] "A prompt-based pipeline for generative game asset synthesis and evaluation," KICS Winter Conference 2026, Pyeongchang, South Korea, pp. 1604–1605 — 2026
Yongtae Lee, ***Junsung Kim***, *and Heung-No Lee*

Honors and Awards

Korean Government Scholarship — 2025.03 – Present
Full tuition and stipend for graduate study at GIST.

2nd Place, The Academic Frontier 2025 — 2025.04
AI Agent & Blockchain Innovation Challenge (idea competition) — Seoul, South Korea
Proposed Planisphere, a decentralized platform for AI-agent prediction competitions, and presented it to the judging panel. Hosted by LECCA Ventures.

Leadership and Outreach

GIST PIUM — 2025.03 – Present
Knowledge-Sharing Volunteer — Gwangju, South Korea
Provide science education and academic and career mentoring to school students, including students from underserved communities, through GIST's student volunteer program.

AI Nights, Ajou University — 2024.09
Organizer and Speaker — Suwon, South Korea
Organized an on-campus AI conference and gave a talk on the K-means and DBSCAN clustering algorithms.

Daesin-Learning, Ajou University 2024.07 – 2024.09
Team Leader Suwon, South Korea
Led a university-supported machine-learning and deep-learning study group; coordinated schedules, budget, and learning materials.

Technical Skills

Programming and frameworks: Python, PyTorch
Tools and environment: Linux (Ubuntu)

Languages

Korean: Native **English:** Fluent

Military Service

National Election Commission 2021.02 – 2022.11
Social Service Agent
Completed mandatory military service as an administrative assistant.

References

Heung-No Lee, Professor
School of EECS, GIST
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M.S. Advisor

Youngin Choi, Ph.D. Student
Department of AI Convergence, GIST
young_in@gm.gist.ac.kr
Senior Lab Mentor